

How does international trade in digital services affect technological innovation?

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Extended abstract

This paper estimates the elasticity of technological innovation with respect to digital services trade in an unbalanced panel of 204 countries observed annually between 2010 and 2022. The count nature of patent applications and the heteroskedasticity of the conditional mean motivate a **Poisson pseudo-maximum likelihood (PPML)** estimator. Identification proceeds through a **Control Function Approach**, which extends instrumental-variable methods to non-linear conditional means by including the first-stage residual in the second-stage regression.

Three instruments support the analysis: (i) a leave-one-out mean of digital services trade across peer countries within k-means clusters formed on eight pre-sample digital infrastructure variables; (ii) a Bartik shift-share interacting time-varying global digitally-deliverable services flows with pre-sample household internet access; and (iii) an analogous Bartik instrument using individual internet use.

Headline result: a one-percent rise in digital services trade raises patent applications by **0.7 to 1.2 percent**, and the over-identifying restrictions are not rejected (Hansen J = 1.06, df = 2, p > 0.5). The marginal innovation response is **increasing** in R&D intensity, financial depth and trade exposure, and **decreasing** in income inequality. Absorptive capacity is the binding margin determining the innovation returns to digital trade integration.

Methodological note: PPML with Control Function

Patent counts are non-negative, right-skewed, and contain country-year observations with zero filings. OLS on the log of patenting drops zeros and is biased under heteroskedasticity in the conditional mean. **PPML** (Santos Silva & Tenreiro, 2006) preserves zeros, allows arbitrary heteroskedasticity, and identifies the same conditional-mean elasticity as a log-linear specification. The conditional mean is $E[TI_{it} | \cdot] = \exp(\alpha + \beta \log DST_{it} + \gamma'X_{it} + \mu_i + \lambda_t)$.

Two-stage least squares is not consistent in non-linear models. The **Control Function Approach** (Wooldridge, 2015) restores identification: in the first stage, OLS regresses $\log(DST)$ on the instruments and the control vector, and the residual $\hat{v}$ is recovered. In the second stage, the PPML regression of patent applications on $\log(DST)$ and the control vector is augmented by $\hat{v}$. The coefficient on $\hat{v}$ provides a *direct test of endogeneity*: its statistical significance in the IV specifications confirms that estimators ignoring endogeneity recover biased magnitudes. Standard errors are clustered at the country level throughout, and the Fuller(4) estimator (Fuller, 1977) is used to dominate 2SLS at moderate first-stage F-statistics.

Note on the ML cluster instrument

The first instrument is a **leave-one-out mean of digital services trade** across peer countries in the same cluster as country i . Clusters are derived from a k-means partition over country averages of **eight digital-infrastructure variables** (broadband subscriptions, mobile penetration, internet users, secure servers, ICT-goods exports, ICT-services exports, R&D expenditure, tertiary education enrolment), computed over the pre-sample window **2005–2009**. Pre-sample weights cannot respond to outcome shocks during 2010–2022; country i is excluded from its own cluster mean. A residual *mirror-feedback risk* via intra-cluster trade linkages remains, mitigated but not eliminated; the Bartik instruments serve as the primary causal anchor.

How the mechanisms work

Three mechanism gradients amplify the DST effect, one attenuates it.

R&D intensity (▲): interaction +0.174***. Higher domestic R&D capacity raises the marginal innovation return to digital trade exposure (Cohen & Levinthal, 1990 — absorptive capacity).

Financial depth (▲): interaction +0.250***. Deeper domestic finance relaxes funding constraints on R&D projects (Hall & Lerner, 2010).

Trade exposure (▲): interaction +2.966***. Pre-existing trade integration complements digital channels in the production of innovation.

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Income inequality (▼): interaction -2.811^{***} . Higher inequality compresses access to the productive units that convert digital-trade gains into patentable knowledge. Absorptive capacity is the binding margin.

Table A1. First-stage results (instrument relevance)

Dependent variable: $\log(DST)$. All specifications include country and year fixed effects.

Instrument	Coefficient	1st-stage F	Conventional?
ML cluster peer mean DST	0.121	9.2	boundary
Bartik 1 (access $\times$ global DST)	0.000***	10.3	yes
Bartik 2 (use $\times$ global DST)	0.000***	9.6	boundary
Joint (3 instruments)	—	9.6	—

Standard errors clustered at country level. Stock–Yogo (2005) threshold for relevance = 10. Bartik instruments are scaled in raw units; their tiny coefficients reflect units, not weakness. Fuller(4) used in all IV specifications. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A2. Mechanism interactions (IV, CFA-PPML, Country & Year FE)

Dependent variable: total patent applications. DST instrumented by ML cluster IV.

Specification	Main effect on $\log(DST)$	Interaction β	1st-stage F
$\times \log(R\&D / GDP)$	3.592***	0.174***	9.3
$\times \log(\text{Broad money} / GDP)$	-1.072***	0.250***	164.5
$\times \log(\text{Trade} / GDP)$	-13.953***	2.966***	241.5
$\times \log(GINI)$	10.715***	-2.811***	1,615

Two coefficients per row: the main effect on $\log(DST)$ and the interaction with each moderator. These are NOT directly comparable on a common axis — the main effect is the conditional elasticity at moderator = 0 (in logs); the interaction captures how the elasticity varies. Marginal effect at any moderator level: $\theta + \delta \cdot M$. Headline mechanism story comes from the interactions; the Bartik-IV main causal elasticity (0.7–1.2, see Table A3) is estimated under a separate specification without interactions.

Table A3. Elasticities by income group (IV, CFA-PPML, Country & Year FE)

Income group	Elasticity (PPML)	1st-stage F (CFA)	Within-country IV reliable?
Low income	-0.060	8,437	yes
Lower-middle income	0.172	310	yes
Upper-middle income	-2.389***	5.8	no (weak)
High income	1.396***	3.7	no (weak)

Pooled PPML with country & year FE and full controls. Within-country IV identification is constrained in the high-income and upper-middle subsamples by weak first stages, so the gradient is best read as descriptive of the conditioning role of national characteristics.

Robustness summary

- **Over-identification:** Hansen J = 1.06 (df = 2), $p > 0.5$ — joint validity of the three instruments is not rejected.
- **Endogeneity test:** the CF residual $\hat{v}$ is significant at the 1% level in both Bartik specifications and in the joint specification, confirming that DST is endogenous in the conditional mean of patenting.
- **Weak-IV diagnostics:** Anderson under-identification statistic = 12.19, Anderson–Rubin Wald = 2.15, Kleibergen–Paap rk Wald F = 4.92 (joint specification). Inference is reported under Fuller(4) (Fuller, 1977) to dominate 2SLS at moderate first-stage F.

Contact: Sergio Martinez-Cotto • DIA Ph.D. Programme • University of St. Gallen. *Comments and methodological feedback are welcomed.* **References:** Akcigit & Melitz (2021); Bartik (1991); Cohen & Levinthal (1990); Fuller (1977); Melitz & Redding (2021); Santos Silva & Tenreyro (2006); Wen, Chen & Zhou (2023); Wooldridge (2015). **Data:** WIPO • UNCTAD Digital Economy Database • World Bank WDI • Penn World Tables 11.0 • V-Dem • WIID.